



Karnataka State Police
Government of Karnataka

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HACKS AT SATELLITE

0atathon 2026

Technology partner



Team Details

Team name: Trinetra Labs

Team leader name: Rayhankhan Pathan

Team size: 1

Problem Statement: PS1 — Intelligent Conversational AI for the KSP Crime Database



Brief about the solution

TRINETRA — Tactical Real-time Intelligence & Network Engine for Tracking, Reasoning & Analytics

An agentic crime-intelligence console for the Karnataka State Police. An investigator asks a question in Kannada or English (voice or text); the AI agent queries the crime database, uncovers hidden criminal networks, profiles repeat offenders, and answers with an interactive graph, hotspot map and timeline — every fact cited to its FIR.

Days → seconds

FIR-to-lead cross-referencing

EN + ಕನ್ನಡ

voice & text, context-aware

100% on Catalyst

deployed, explainable, secure



Opportunities

- a. How different is it from any of the other existing ideas?
- b. How will it be able to solve the problem?
- c. USP of the proposed solution

How it's different

- Not a dashboard and not a plain chatbot — an agent that returns understanding.
- Generative UI: it decides whether to draw a network, a map or a timeline.
- Relationships modelled as edge tables → graph in code (no graph DB needed).
- Kannada + English voice, built for Indian policing.

How it solves the problem

- Natural language → safe ZCQL over the Data Store, no SQL skills needed.
- Surfaces hidden offender / phone / money links across jurisdictions.
- Transparent risk scoring prioritises investigation.
- Cuts manual cross-referencing from days to seconds.

USP

- Agentic + explainable: every answer cites its FIRs (evidence trail).
- Spatiotemporal hotspots + organised-network detection in one console.
- Fully on Zoho Catalyst — Qwen LLM, Zia voice, Data Store, Slate.
- Production-grade: role-based access, audit-ready, scalable.



List of features offered by the solution

Conversational agent

Voice + text, English & Kannada, context-aware follow-ups.

Criminal network analysis

Hidden links between offenders, victims, phones, accounts.

Spatiotemporal hotspots

District clusters layered with time-of-day.

Money-trail analysis

Suspicious transactions and mule-account links.

Crime forecasting

Projected next-week hotspots for proactive patrol.

Explainable & cited

Shows the generated ZCQL + reasoning; cites FIRs; PDF export.

NL → ZCQL retrieval

Plain questions become safe, validated Data Store queries.

Offender risk profiling

Explainable 0–100 score from weighted factors.

Investigator timeline

Auto case chronology showing MO escalation.

Similar-case leads

Find look-alike cases by MO — investigative leads.

Early-warning alerts

Spikes, forming rings, repeat offenders.

Secure & governed

Role-based access, audit logs, SELECT-only query gate.



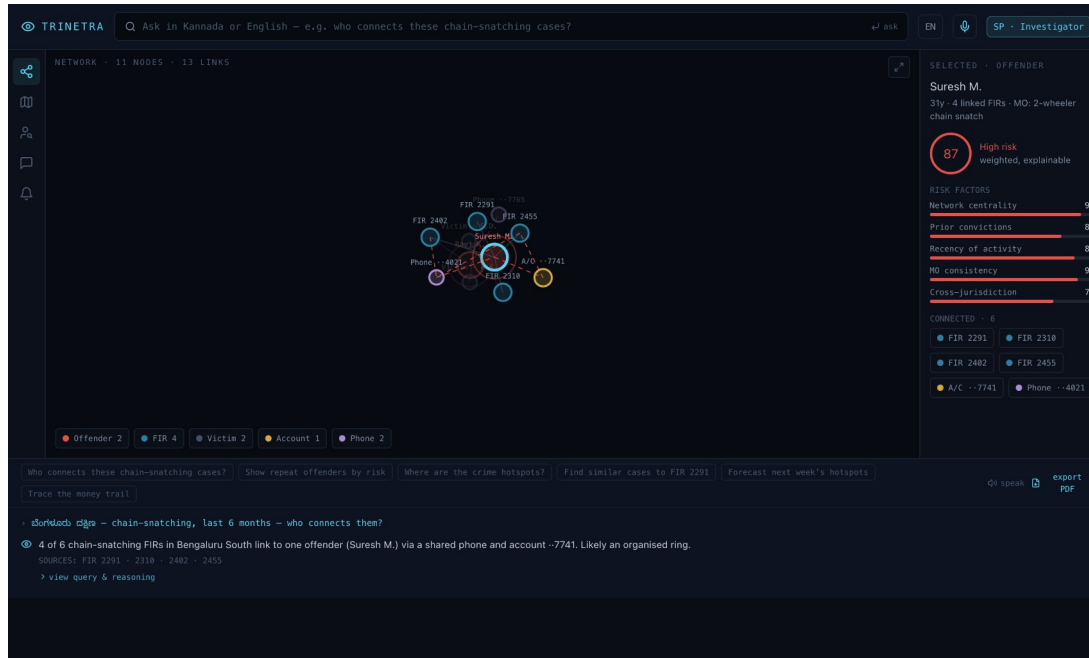
Process flow diagram or Use-case diagram



Use case: “Who connects these chain-snatching cases?” → the agent runs the query, finds 4 FIRs tied to one offender via a shared phone + mule account, and renders the ring with an evidence trail — in seconds.



Wireframes/Mock diagrams of the proposed solution (optional)

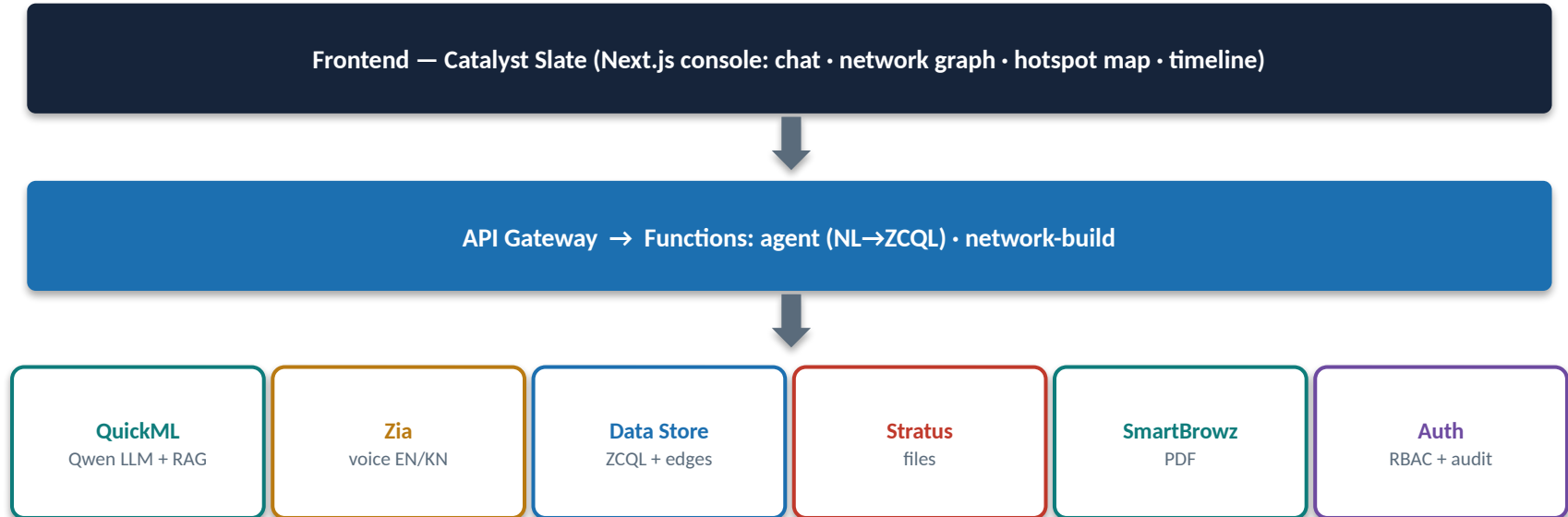


Console layout

- **Command bar** — natural-language + voice query
- **Network canvas** — force-directed graph, neighbour highlight
- **Inspector** — risk factors + connected entities
- **Agent log** — cited answer · view query & reasoning · PDF



Architecture diagram of the proposed solution



Synthetic data → Data Store. Every generated ZCQL is validated (SELECT-only, whitelisted tables) before execution — the governance gate.



Technologies to be used in the solution

Frontend

Next.js 14 · React · TypeScript · Tailwind CSS · react-force-graph-2d · MapLibre GL · Web Speech API

Backend / Functions

Node.js Catalyst Functions (Advanced I/O) · ZCQL · REST + API Gateway

AI / ML

Qwen via Catalyst QuickML LLM-Serving · RAG knowledge base · Zia STT / TTS / translation

Data

Catalyst Data Store (relational + edge tables) · Python synthetic-data generator (planted rings)

Platform

Zoho Catalyst — Slate hosting · Authentication · Stratus · SmartBrowz · Cron · Signals



List down Catalyst Services being used in the solution

Slate / Web Hosting

Hosts the Next.js console

Functions (Advanced I/O)

Agent (NL→ZCQL) & network-build logic

API Gateway

Routing, throttling, auth in front of Functions

QuickML — LLM Serving + RAG

Qwen for NL→ZCQL & grounded answers

Zia Services

Kannada/English speech-to-text, TTS, translation

Data Store (ZCQL)

Crime DB + relationship edge tables + search

Stratus

Case files, evidence, exported PDFs

SmartBrowz

Conversation → PDF report generation

Authentication

Role-based access (investigator/analyst/supervisor)

Cron + Signals

Scheduled risk recompute & early-warning alerts



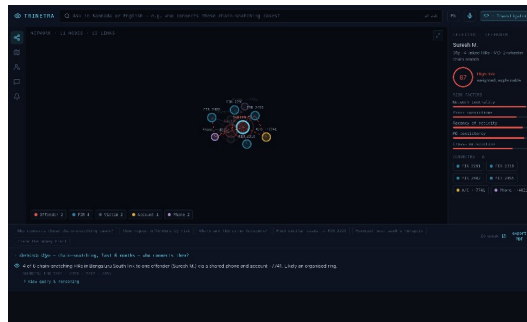
Estimated implementation cost (optional)

Built within the hackathon Catalyst credits (₹1,500 + \$250). Indicative monthly run-rate at SCRB pilot scale:

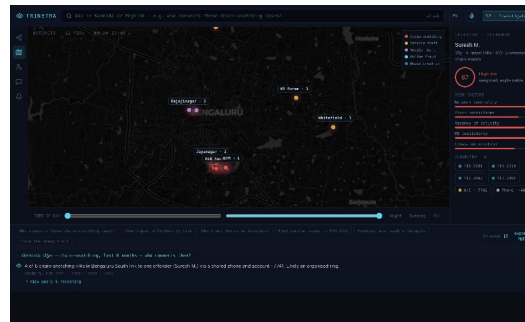
Catalyst service	Usage	Indicative / month
Functions (agent + network)	~50k invocations	Within free tier → low
Data Store	~2-5 lakh records	Low
QuickML LLM-Serving (Qwen)	~30k queries	Usage-based
Zia voice	~5k minutes	Usage-based
Slate + Stratus + SmartBrowz	hosting + PDFs	Low

Snapshots of the prototype

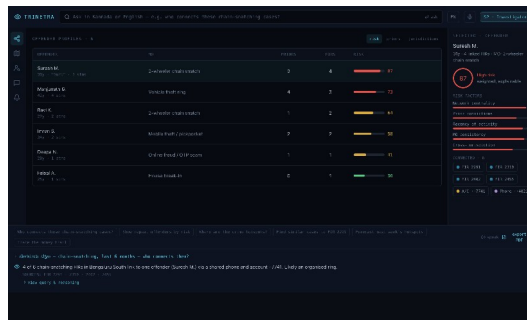
Network — organised ring



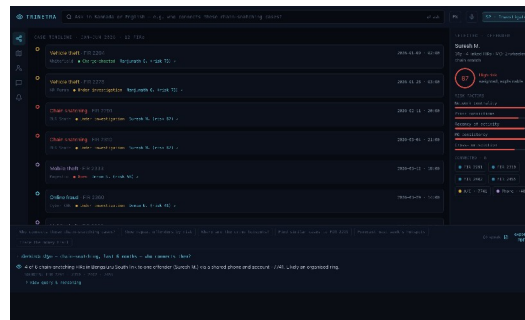
City hotspot map (MapLibre)



Offender risk profiles



Case timeline





Prototype Performance report/Benchmarking

< 1s

agent response (mock mode)

100%

malicious queries blocked by ZCQL gate

1.5 lakh

FIRs generated & queried in tests

6/6

demo intents return grounded, cited answers

16 nodes

network built per offender ego-graph

11 tables

relational schema, edge-modelled



Provide links to your:

1. **GitHub Public Repository**
2. **Demo Video Link (3 Minutes)**
3. **Deployed Link**

GitHub repository: <https://github.com/mrayhankhan/trinetra>

Demo video (3 min): <https://www.youtube.com/watch?v=UL8YjVzIKK0>

Deployed link (Catalyst): <https://trinetra-60074156791.development.catalystserverless.in/app/index.html>



Additional Details/Future Development (if any)

RAG criminology knowledge base — Ground answers in IPC / case-law / SOP documents via QuickML RAG.

Crime forecasting — QuickML time-series + Zia AutoML to predict emerging hotspots.

Financial-crime expansion — Deeper money-mule and cross-bank transaction graphs.

More Indic languages — Beyond Kannada — Hindi, Tamil, Telugu, Urdu.

Mobile + field app — Push alerts and on-the-go queries for officers.

Live SCRB integration — Connect to real KSP systems with full governance & DPDP compliance.



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THANK YOU